← Back **Week 6 quiz**
Graded Assignment • 20 min

Due Mar 18, 11:59 PM CET

1. In the lectures, you've learnt our Universe has both a normal and Dark side! Can you choose the option that specifies the percentages and nature of the different components our Universe is made up of? 1 point

- ○ 50% normal matter, 25% Dark matter that tries to pull things together through gravity and 25% dark energy that is expanding the Universe at an accelerating rate.
- ○ 5% normal matter, 25% Dark matter that tries to pull things together through gravity and 70% dark energy that is expanding the Universe at a decelerating rate.
- ◉ 5% normal matter, 25% Dark matter that tries to pull things together through gravity and 70% dark energy that is expanding the Universe at an accelerating rate.
- ○ 25% normal matter, 25% Dark matter that tries to pull things together through gravity and 50% dark energy that is expanding the Universe at an accelerating rate.

2. Andy told us how important spectroscopy is to study astrophysical objects. Of the choices given, pick all that describe the advantages spectroscopy can provide. 1 point

- [x] Spectroscopy can be used to measure the velocity an object is moving with.
- [ ] Spectroscopy is the best way to find Supernovae.
- [ ] Spectroscopy can be done faster than imaging.
- [x] Spectroscopy can be used to infer the chemical composition or make-up of an object.

3. Catherine told us about the wonderful Supernovae Ia (SN Ia) that are called "standard candles" and used to measure distances in cosmology. What is the reason that these objects are standard candles? 1 point

- ◉ SNe Ia form when a white dwarf explodes on reaching a critical mass of 1.4 solar masses. Since all SNe Ia explode at the same mass, they produce the same luminosity, marking them as "standard" candles.
- ○ SNe 1a have the same apparent brightness no matter how far away they are from us - this is why they are called standard candles.
- ○ SNe Ia form when a red giant explodes on reaching a critical mass of 1.4 solar masses. Since all SNe Ia explode at the same mass, they produce the same luminosity, marking them as "standard" candles.
- ○ Every star can explode as a SN Ia - this is why they are called standard candles.

4. Doing spectroscopy of a large number of objects is really slow. The best way to improve this is by: (pick one answer) 1 point

- ○ using bigger telescopes
- ○ integrating/looking at a galaxy for longer times
- ○ integrating/looking at a galaxy for shorter times
- ◉ using multiple fibres to look at many sources at the same time.

5. All star-forming galaxies produce a Lyman Alpha Emission line at a wavelength $\lambda_{emit} = 121.6$ nm when the electron in a Hydrogen atom falls from the first excited state to the ground state. The energy produced by this transition is E = 10.2 eV. Imagine a galaxy at a redshift of z=9 that produces a Lyman Alpha line. As an observer on earth, what observed wavelength (λ_{obs}) and energy would we measure for this line? Remember $E = \frac{hc}{\lambda}$. 1 point

- ◉ $\lambda_{obs} = 1216$ nm and E = 1.02 eV
- ○ $\lambda_{obs} = 12160$ nm and E = 1.02 eV.
- ○ $\lambda_{obs} = 12160$ nm and E = 10.2 eV.
- ○ $\lambda_{obs} = 12160$ nm and E = 100.2 eV.